Intelligent Multi-Agent Rainwater Management for Urban Flood Mitigation

CS3081 – Report 1: Project Description & Problem Statement

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1. Project Description

This project proposes a simulated multi-agent system, intended for urban planners and emergency-management teams, that manages rainwater during a storm in a synthetic urban district modelled on Jeddah. The environment is a graph of zones — roads, residential areas and storage basins — linked by drainage paths; the current design uses two storage basins. Five agents coordinate through a shared environment model holding each zone's water level, drainage connections and status:

- **Rainfall Monitoring** estimates rainfall and flags zones approaching capacity each time step, using probabilistic reasoning to account for noisy or conflicting sensor readings.
- **Water Collection** uses rule-based logic (e.g. "basin above 90% full: refuse inflow") to decide where excess water should be directed.
- **Route/Drainage** computes A* paths through the drainage network, avoiding blocked or high-risk connections, and replans when a route becomes unusable.
- **Storage Management** monitors each storage basin's fill level and reallocates incoming flow to an alternative basin when one nears capacity.
- **Emergency Response** applies rule-based prioritisation to decide which at-risk zones (e.g. a hospital access road) are addressed first.

Only classical AI techniques from the course are used; no machine learning.

2. Problem Statement

Jeddah lies on a coastal plain below mountains, so heavy rain produces flash floods along wadi pathways through built-up areas [1]. The floods of November 2009 and January 2011, with rainfall of 70 mm and 111 mm over three hours each, killed 113 people in 2009 and damaged more than 10,000 homes and 17,000 vehicles [1]. On 24 November 2022 the National Center for Meteorology recorded 179 mm in six hours, the highest ever recorded in the city [2]. Because extreme rainfall can recur, flood response must work during the event itself, not only through long-term infrastructure.

Real storms create the adversarial conditions this course studies: sensors give unreliable readings, roads become blocked unexpectedly, and several reservoirs compete for the same inflow. A fixed drainage plan cannot adapt to these changes, and a multi-agent design suits the problem because these decisions are distributed and must be coordinated under partial information. The research question is therefore: *how can a team of agents reroute and store rainwater correctly, as measured by Objectives 1–3, when knowledge is incomplete, observations are noisy and conditions change mid-event?*

3. Project Objectives

Scenarios run for 120 steps at light (10 mm/h), heavy (30 mm/h) and extreme (60 mm/h) rainfall. A zone is flooded when its water level exceeds its capacity. The baseline uses fixed shortest-path drainage routes computed once, with no communication or replanning.

1. **Reduce flooding.** Reduce mean peak flooded area by at least 30% versus the baseline in every scenario where the baseline floods, averaged over 30 runs per scenario.
2. **Capture runoff.** Store at least 60% of total runoff volume (stored ÷ generated) in the heavy scenario, with total reservoir capacity set to 75% of that runoff, and no reservoir overflowing in more than 10% of runs.
3. **Adapt to disruption.** In at least 45 of 50 randomised runs (90%), replan within 5 time steps of every injected fault: 20% sensor dropout, Gaussian noise ($\sigma = 5$ mm/h) and 1–3 drainage connection closures. A replan succeeds when the new route is valid and avoids all closed connections.

4. Expected Outcome

The expected outcome is a working, visual simulation that demonstrates coordinated agent behaviour under uncertainty. Experiments will compare the multi-agent system against the baseline on four metrics: water collected vs. lost, flooded area extent, storage utilisation and adaptation success rate, and will serve as the base for the later reports.

References

[1] A. M. Youssef, S. A. Sefry, B. Pradhan, and E. A. Alfadail, "Analysis on causes of flash flood in Jeddah city (Kingdom of Saudi Arabia) of 2009 and 2011 using multi-sensor remote sensing data and GIS," Geomatics, Natural Hazards and Risk, vol. 7, no. 3, pp. 1018–1042, 2016.

[2] "Jeddah receives 179mm of rainfall, higher figure than 2009 peak," Arab News, Nov. 25, 2022. [Online]. Available: https://arabnews.com/node/2205676. Accessed: Nov. 9, 2026.

Appendix: AI Use Declaration

(a) AI evaluation summary and score

Evaluation of the original draft by Claude acting as a strict professor (table condensed from the full output). **Score: 6.0 / 10.**

Criterion	Score	Strengths	Weaknesses	Recommended fix
Problem clarity (3)	2.0 / 3	Grounded in a real, cited Jeddah context; the research question names the three difficulties.	"Rainfall intensity is rising" unsupported (different durations, three data points); "drainage is still limited" uncited; "adversarial" flagged as the wrong term; "correctly" undefined; no justification for a multi-agent approach.	Qualify or remove the trend claim; verify the 70 mm figure; replace "adversarial"; define "correctly"; add one sentence justifying a multi-agent design.
Project description (2)	1.0 / 2	Names all five agents and the techniques; scope is clear (no ML).	Reads as a list of names: no agent roles, no environment description, no communication mechanism, no technique-to-agent mapping; "planning" unspecified.	One line per agent (role and technique) and one sentence describing the environment.
Objectives and expected outcome (3)	1.5 / 3	Numeric targets, a baseline, scenarios, and metrics that map to the objectives.	"Set number of time steps" never given; flooded area, replan success and collected vs. lost undefined; scenarios and fault parameters undefined; targets unjustified; Objective 2 bundles two goals; baseline and run counts unspecified.	Define every metric and threshold; specify baseline, scenarios, number of runs and fault injection.
Organization and academic writing (2)	1.5 / 2	Clear numbered structure, concise, IEEE-style references.	Formatting artifacts (stray bold markers); rhetorical phrasing; [2] is a news article for a meteorological figure with no access date; outcome paragraph repeats the objectives.	Clean formatting; cite a primary source or add an access date; shorten the outcome paragraph.
Total	**6.0 / 10**			

(b) Changes made in response

1. Made all three objectives testable by defining the scenarios (10/30/60 mm/h), flooded-cell threshold (0.15 m), baseline, 30 runs per scenario, the 5-step replanning bound, and the fault parameters; the vague "balanced storage" goal became a measurable overflow limit.

2. Replaced the list of agent names with one line per agent giving its role and technique, and added the environment (20×20 grid, road graph, four reservoirs) and the communication mechanism (shared blackboard).

3. Removed the unsupported claim that rainfall intensity is rising and drainage is limited; corrected the 2022 figure to 179 mm in six hours, as reported, rather than "a single day"; and gave the 2009/2011 rainfall as 70 mm and 111 mm over three hours each, with the 2009 death toll of 113, as confirmed against [1], so the durations are comparable.

4. Defined "correctly" in the research question through the objectives and added a sentence justifying a multi-agent design. I did not apply the AI's suggestion to drop "adversarial", because it is this course's own term; I defined it instead.

5. Removed formatting artifacts, added an access date to reference [2], and shortened the Expected Outcome by deleting its repeated closing sentence.

6. Added a phrase to the Project Description naming the intended users (urban planners and emergency-management teams), per the report requirements.

7. Replaced the environment description (originally a 20×20 grid with four reservoirs and a formal blackboard with Bayesian sensor fusion) with the architecture the team actually designed and has begun implementing: a graph of zones connected by drainage paths, two storage basins, A* route planning with replanning around blocked edges, and rule-based storage reallocation — so the report reflects the system as designed and coded rather than a more elaborate alternative.

(c) Verification log

Checks run while revising (all items checked by the AI against web sources or by the student against the cited paper):

✓ 179 mm on 24 Nov 2022, recorded by the National Center for Meteorology as the highest ever in Jeddah: matches Arab News, Saudi Gazette and Al Arabiya reports found by web search.

✓ The 179 mm fell between 8 a.m. and 2 p.m. (six hours), not over a full day: matches Saudi Gazette, Arab News and Watchers; the report wording was changed accordingly.

✓ 70 mm (2009) and 111 mm (2011) over three hours each, 113 deaths in 2009, and more than 10,000 homes and 17,000 vehicles damaged: confirmed by the student against the text of [1]; the AI did not open [1] itself.

✓ Coastal-plain location and wadi runoff causing flash floods in urban areas: confirmed by the student against the text of [1].

✓ “Drainage is still limited” and “rainfall intensity is rising”: no supporting source found, so both claims were removed.

✓ Simulation parameters (grid size, rainfall rates, thresholds, run counts) are design choices, not facts; the team has agreed to them and they must match the implementation.

(d) AI tools used

I used Claude (Anthropic) in one chat to generate the first draft of the report, and in a second chat to evaluate it as a strict professor and to revise the report and appendix in response.